

HR Analytics: Boosting Retention with Data Insights at Adecco India

Background

Adecco India, a medium-sized technology company specializing in software development, has a diverse workforce spanning departments such as engineering, marketing, sales, and customer support. Known for its innovative solutions and client-centric approach, the company has achieved steady growth in recent years. However, a growing concern over employee turnover, especially among junior-level sales employees, has prompted management to prioritize retention strategies.

Employee attrition not only disrupts team performance but also incurs significant costs related to recruitment, onboarding, and training. Recognizing the importance of employee satisfaction and engagement in achieving long-term organizational goals, Adecco India has decided to leverage HR analytics to address this challenge.

The company is committed to using data-driven insights to understand the root causes of attrition and job dissatisfaction, enabling the development of effective strategies to boost employee retention and foster a supportive work environment.

Problem

Adecco India is grappling with a high employee attrition rate, particularly among junior-level sales employees. This issue is disrupting team dynamics, reducing productivity, and increasing operational costs associated with recruitment, onboarding, and training of new hires.

The problem has broader implications for the company, as frequent turnover adversely impacts overall performance, employee morale, and the ability to achieve long-term organizational goals.

Addressing this issue is essential to:

1. **Minimize Costs:** Reduce expenses tied to recruitment and training.
2. **Enhance Productivity:** Maintain consistent team performance by fostering employee stability.
3. **Improve Workforce Morale:** Build a motivated and engaged workforce that supports organizational success.

Project Scope

This project focuses on analyzing HR data to identify factors contributing to high employee attrition at Adecco India, particularly among junior-level sales employees. By leveraging insights from employee demographics, job satisfaction, performance ratings, and work-life balance metrics, the project aims to:

1. **Understand Attrition Patterns:** Analyze trends and key drivers of employee turnover.
2. **Identify High-Risk Areas:** Highlight departments, job roles, or demographics with elevated attrition rates.
3. **Develop Retention Strategies:** Provide actionable recommendations to improve employee satisfaction, engagement, and retention.
4. **Support Strategic Decisions:** Enable data-driven decision-making for HR policies and workforce management.

Data Overview

The dataset used for this project is sourced from Adecco India's HR records, which includes detailed employee information across various departments. Key variables in the dataset include:

1. **Employee Demographics:** Age, gender, education, marital status, and distance from home.
2. **Job Information:** Department, job role, job level, years at the company, performance ratings, and job satisfaction.
3. **Work Environment:** Work-life balance, environment satisfaction, and involvement.
4. **Compensation and Benefits:** Monthly income, daily rate, stock options, and overtime status.
5. **Attrition Data:** Whether the employee has left the company (attrition status).

The data also includes employee feedback, such as satisfaction with relationships and the work environment, as well as training and performance metrics. This rich dataset allows for a comprehensive analysis of factors contributing to employee attrition.

Methodology

Data sources

1. SQL
2. AWS
3. Data Scraping
4. Local data sources
- **Data wrangling**
 1. Data understanding
 2. Data cleaning
 3. Data merging and joining
 4. Data manipulation
- **Data analysis**
 1. Finding the trends and patterns

- **Data visualisation**

Dashboard Overview

This HR Attrition Dashboard provides comprehensive insights into employee attrition, highlighting key trends and actionable points for improvement. Here's a summary based on your provided data:

Overview

- **Total Employees:** 1,470
- **Attrition Rate:** 17% (237 employees left)
- **Active Employees:** 1,233

Attrition by Job Role

- The highest attrition rates are observed in **Sales Executive**, **Research Scientist**, and **Laboratory Technician** roles. These roles account for the largest percentage of overall attrition.

Attrition by Field

- **Life Sciences** and **Medical** fields show the highest attrition, with **89** and **63** employees leaving, respectively.

Attrition by Business Travel

- Employees who travel **rarely** or **frequently** show higher attrition, indicating that travel frequency may correlate with job dissatisfaction or burnout.

Attrition by Age Group and Gender

- The data segmented by **age group** and **gender** highlights that younger employees (especially those in the **48-53** age group) have a lower attrition rate. However, **male employees** have a higher overall attrition rate than female employees.

Stakeholders

Internal Stakeholders:

1. **HR Department:** Utilizes the dashboard to track attrition trends, identify patterns, and devise retention strategies.

2. **Sales Department:** Reviews attrition data to understand employee turnover and improve sales team retention.
3. **Engineering Department:** Analyzes attrition trends to retain technical staff and address issues specific to the department.
4. **Marketing Department:** Uses the data to identify turnover in marketing roles and work with HR to improve engagement and retention.
5. **Customer Support Department:** Monitors attrition rates in customer support to maintain a skilled team and address retention concerns.
6. **Senior Management:** Leverages insights from the dashboard to make strategic decisions that improve overall employee satisfaction and retention.

External Stakeholders:

1. **Recruitment Agencies:** Uses attrition data to help HR teams with talent acquisition and ensure alignment with company needs.
2. **Training Providers:** Reviews attrition patterns to suggest training programs that enhance employee satisfaction and reduce turnover.

Technical Processes

- Use pivot tables for summarizing data.
- Calculate averages, variances, and growth rates.
- Create charts and graphs for visual representation.
- Apply filters and sorting for specific analyses.
- Use functions like Count and Sum IF for data aggregation.

Key Findings

1. **Overall Attrition:**
 - **Total Employees:** 1,470
 - **Attrition Rate:** 237 (17%)
 - **Active Employees:** 1,233
2. **Attrition by Job Role:**
 - **Highest Attrition:** Sales Executive (57), Research Scientist (47), Laboratory Technician (62)
 - **Lowest Attrition:** Research Director (2), Manager (5)
3. **Attrition by Employee Field:**
 - **Highest Attrition:** Life Sciences (89), Medical (63)
 - **Lowest Attrition:** Human Resources (7)
4. **Attrition by Business Travel:**
 - **Non-Travel:** 12

- **Frequent Travel:** 69
- **Rare Travel:** 156
- 5. **Attrition by Age Group and Gender:**
 - **Male:** 63.29% attrition rate
 - **Female:** 36.71% attrition rate
 - **Average Age:** 37 years
- 6. **Attrition by Environment Satisfaction:**
 - **Very Dissatisfied:** 72
 - **Dissatisfied:** 62
 - **Neutral:** 60
 - **Very Satisfied:** 26

7. Attrition by Marital Status:

- **Divorced:** 33 employees (13.9% of total attrition)
- **Married:** 84 employees (35.4% of total attrition)
- **Single:** 120 employees (50.6% of total attrition)

Grand Total: 237 employees (100% of total attrition)

8. Attrition by Overtime:

- **No Overtime:** 944 employees (76.5% of total employees)
- **Yes Overtime:** 289 employees (23.5% of total employees)

Recommended Analysis

1. **Job Role Attrition:** Investigate high attrition in Sales Executive, Research Scientist, and Laboratory Technician roles. Understand challenges and improve retention strategies.
2. **Business Travel:** Analyze the high attrition rate among employees who travel frequently. Improve travel policies and work-life balance.
3. **Marital Status:** Focus on single employees with the highest attrition. Offer support and work flexibility.
4. **Environment Satisfaction:** Address dissatisfaction in the work environment. Improve workspace, management support, and team dynamics.
5. **Age Group:** Explore why older employees (48-53) have lower attrition and replicate successful retention strategies in younger employees.
6. **Overtime:** Examine the link between overtime and higher attrition. Introduce measures to reduce burnout and improve work-life balance.

7. **Performance Ratings:** Assess if performance ratings correlate with attrition. Target interventions for low performers and high performers feeling undervalued.
8. **Field-Specific Retention:** Tailor retention strategies for Life Sciences and Medical fields to enhance job satisfaction and reduce attrition.

Conclusion:

The HR Attrition Analysis at Adecco India has provided valuable insights into the key factors contributing to employee turnover, particularly in high attrition job roles, fields, and specific employee demographics. Key findings highlight that Sales Executives, Research Scientists, and employees in Life Sciences and Medical fields face higher attrition rates. Additionally, frequent business travel, overtime, and dissatisfaction with the work environment contribute to higher turnover.

Based on these findings, targeted retention strategies should focus on improving work-life balance, addressing job-specific challenges, offering career growth opportunities, and enhancing employee satisfaction in key areas. It is crucial to create a positive work environment, refine business travel policies, and provide tailored support to employees based on their job roles and needs.

Ultimately, by implementing these data-driven insights and strategies, Adecco India can work towards reducing attrition rates, improving employee satisfaction, and fostering a more stable and motivated workforce.

Project owner

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